

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Bachelor of Software Engineering (Honours)

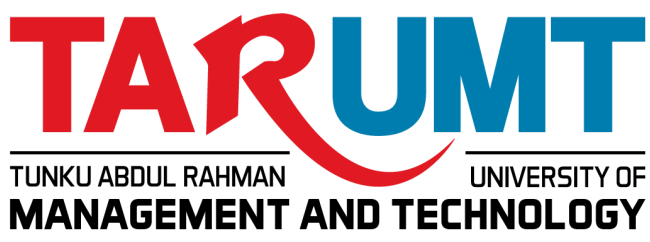
Programme: RSW (Group: xxxxxx)

ASSIGNMENT

BMSE2073 SOFTWARE DESIGN AND ARCHITECTURE

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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
Plagiarism Statement and Guideline for Late Submission of Coursework

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted work is all our work and is in our own words.

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SOFTWARE DESIGN DOCUMENT

**PROJECT TITLE : Digital Transformation for Local
Artisans & Craftspeople**

PRACTICAL GROUP : RSW G5

NAME	MODULE
1. ONG WENG KANG	Social Media Integration
2. CHANG KAR XI	Storefront Management
3. KENNARD KHOO TENG PHIN	Payment Gateway Integration
4. TAN GUAN HONG	AI Recommendation Engine

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1. Introduction

1.1 Document Purposes

This document shall serve as :

a. Formal design reference for developers

- Offering precise blueprints of system architecture, design decisions, modules, interfaces, and data structures so that the development teams can implement the system consistently and correctly.

b. Communication artefact for stakeholders

- Communicate the system scope, architecture, quality objectives, and UI/UX sketches to project stakeholders (project manager, course instructors, crafts representatives, and testers) so that they understand the planned solutions and can provide feedback.

c. Test and verification guide

- Provide the information needed to design module-level and integration test cases (activity flow, class responsibility, design pattern, and interface contract) to verify whether the system meets functional and non-functional requirements.

d. Maintenance and future evolution guide

- Document the principles and limitations behind the main architecture and pattern choices so that future maintainers can understand why things should be done a certain way and be able to safely scale the system.

e. Assessment deliverables

- To meet course assignment requirements, provide documented deliverables, such as architecture diagrams, quality objective justifications, activity diagrams, class diagrams with design patterns, and UI models as required in the module descriptions.

1.2 Project Summary

Project Title: Digital Transformation for Local Artisans & Craftspeople (ArtisanMarket)

Problem Statement

Many local artisans lack convenient online sales channels, digital marketing skills, and data analytics tools, making it difficult to grow their small handicraft businesses. This platform aims to fill these gaps, helping artisans register, create shops, list products, learn digital skills through training modules, and gain support in data analytics and marketing.

Project Objective

- Provide artisans with an easy-to-use platform, enabling them to create and manage online stores through a simple registration process (Usability).
- Allow buyers to browse, purchase, and rate artisan products, with secure payment processing (Functionality & Security).
- Provide training materials and workshops to help artisans improve their product presentation and digital marketing skills (Education/Adoption).
- Provide basic sales analytics and automated marketing suggestions to help artisans increase product visibility and sales (Support & Maintainability).
- Support multi-currency and region-based shipping options to expand local/regional business (Actual Coverage).

Primary Users & Roles:

- **Artisans / Sellers** — register, manage storefronts, upload products, view orders, access training.
- **Buyers** — browse catalogue, place orders, manage payments and returns.
- **Trainers / Admins** — upload training content, moderate platform content, manage artisan approvals.
- **System Administrator** — system configuration, manage integrations (payment, shipping), platform maintenance. (not sure)

System Modules

1. Storefront Management

This module allows artisans to create and manage their online storefronts.

Key features include:

- Upload product images
- Set product descriptions and pricing
- Update stock/inventory levels
- Manage product categories
- Publish or unpublish products

2. Payment Gateway Integration

This module handles secure online transactions between buyers and artisans.

Key features include:

- Secure checkout process
- Support for multiple currencies
- Integration with third-party payment providers (e.g., FPX, e-wallets, credit/debit card processors)
- Transaction validation and payment status updates

3. AI Recommendation Engine

This module enhances user experience and business growth using AI-driven insights.

Key features include:

- Personalized product recommendations for buyers
- Suggested marketing strategies for artisans
- Analysis of browsing and purchase patterns
- Product visibility optimization

4. Social Media Integration

This module helps artisans promote their work automatically.

Key features include:

- Auto-posting new product listings to social media platforms
- Generating promotional captions and hashtags
- Sharing artisan stories or achievements
- Increasing product reach and brand visibility

5. Learning & Workshop Portal

This module provides digital training resources for artisans to improve their business skills.

Key features include:

- Tutorials on product photography, pricing strategy, branding, and customer engagement
- Pre-recorded lessons and reading materials
- Live webinars or online workshops
- Progress tracking and completion badges/certificates

1.3 Document Overview

This SDD is organised to match the course template and contains the following chapters:

Chapter	Title	Contents Summary
Chapter 1	Introduction	Purpose of this document, project summary, scope, stakeholders, success criteria.
Chapter 2	Initial Design	Architecture diagram, selected Quality Objectives, three candidate architectural patterns with brief pros/cons, final pattern selection and justification. (Critical section — maps to assignment requirements to choose and justify architecture.)
Chapter 3	Detailed Design	Module-based designs. For each team member / module: activity diagrams for assigned use cases, design class diagram implementing one design pattern, rationale for chosen pattern.
Chapter 4	User Interface / UX Design	Minimum two mockup pages per module (Figma links or embedded images) showing cohesive UI flows and style notes (typography, spacing, accessibility considerations).
Chapter 5	Appendices	Glossary, data entity list, sequence diagrams (optional), references, and any external API contract samples (payment/shipping).

2. INITIAL DESIGN

2.1 Architectural Design

Chapter 5

Note:

- ***Draw your selected architecture diagram*** for the project in this section
- *Each structure/segment/layer/section must include the system's main components.*
- *A good architecture will show how components in each module are arranged and communicated in the architecture diagram.*
- *Relationship*

2.2 Design Rationale For The Architecture

2.2.1 Quality Objectives

2.2.1.1 Scalability

Scalability is a fundamental quality objective for digital transformation platforms, as the system must support the continuous growth of users and digital products. With more artisans joining the platform across Malaysia, the number of product listings, uploaded images, training modules, and customer interactions will steadily increase. Buyers will need to browse a wide variety of handcrafted items, engage with multimedia content, and conduct frequent searches. This will place significant computational and storage demands on the system. Without a scalable architecture, this increase in load will lead to slow response times, longer loading times, and system instability, ultimately negatively impacting the user experience. To maintain the platform's competitiveness and availability, the system must be able to scale seamlessly horizontally and vertically with fluctuating workloads.

Scalability is **particularly critical due to the presence of AI-driven personalization and recommendation services** within the platform. AI algorithms typically require powerful processing capabilities, especially when analyzing buyer behavior, filtering products, or generating real-time handcrafted item recommendations. During peak periods, hundreds of buyers may trigger recommendation queries simultaneously, potentially creating bottlenecks if the system lacks the ability to scale individual components. Furthermore, data-intensive processes such as image recognition, product categorization, and keyword extraction will further increase computational demands. To address these challenges, the architecture must support the independent scaling of microservices, allowing the AI engine, product search, and media processing modules to allocate additional resources as needed without impacting other parts of the system.

Seasonal peaks in user activity are another crucial reason for prioritizing scalability. Cultural holidays such as Hari Raya Aidilfitri, Deepavali, Chinese New Year, and Christmas are critical sales periods for artisans, as customers actively seek out traditional and handcrafted items. During these periods, online visitors and transaction volume surge dramatically in a short time. Artisans may upload new batches of products, while buyers engage in numerous searches, add items to their carts, and complete purchases. If the system doesn't dynamically scale for these peak periods, it may be unable to handle the sudden load, leading to checkout failures, lost images, and website inaccessibility. These outages directly impact artisans' revenue and damage the platform's reputation. Therefore, scalability ensures the platform remains stable even when demand temporarily exceeds normal traffic levels.

Furthermore, scalability **supports the platform's long-term vision** as a sustainable digital ecosystem, rather than a short-term prototype. The platform aims to expand its functionality beyond simple e-commerce, integrating training workshops, virtual craft fairs, collaboration tools, and potentially expanding into international markets. New modules may require additional processing power or a separate deployment environment. Scalability ensures that the system can integrate these future enhancements without requiring major architectural refactoring. By designing the system to be scalable from the outset, developers have ensured that the platform is future-proof, supports Malaysia's growing artisan community, and can handle increasing usage nationwide without compromising performance or reliability.

2.2.1.2 Maintainability

Maintainability is a critical quality objective because the platform integrates numerous internal modules and third-party services, all of which are likely to be frequently updated and iterated. The system includes functions such as user management, product management, workshop scheduling, payment processing, AI inference, and social media integration. These components update at varying frequencies and involve various external dependencies. For example, payment gateways like FPX or Stripe regularly update their API requirements and security policies, while social media platforms frequently modify their authentication rules and content sharing mechanisms. A highly maintainable system ensures that these changes can be managed smoothly without extensive code rewriting or system downtime. This is especially important because the platform needs to operate continuously, serving artisans and buyers.

Maintainability also **supports software engineering workflows**, enabling different development teams or student contributors to develop different modules independently without conflicts. As part of the project scope, new features such as virtual craft exhibitions, artisan rating systems, or integrated logistics services may be added in subsequent development cycles. Without maintainability, adding new features risks breaking existing functionality, slowing development progress, or causing regressions. Therefore, the architecture must promote loose coupling, clear separation of concerns, and modularity, allowing developers to update individual modules independently. This independence reduces the likelihood of unexpected side effects and lowers the overall cost of updating or fixing bugs.

Furthermore, maintainability directly **impacts user experience and platform reliability**. When issues arise, such as incorrect product information, incorrect workshop schedules, or failed social media postings, the technical team must be able to quickly identify and resolve the problems. A maintainable architecture provides clear module boundaries, simpler debugging paths, and predictable behavior. This ensures faster fault recovery and reduces the risk of prolonged downtime. For artisans who rely on the platform to promote and sell their work, any downtime directly impacts their income; therefore, maintainability is crucial for maintaining user trust and platform stability.

Finally, maintainability plays a critical role in the **long-term maintenance of the platform**, especially considering that student teams or rotating developers may continue to improve the project after the initial deployment. A maintainable system ensures that new contributors can quickly understand the architecture and confidently make modifications. As technology evolves, certain modules, especially those involving artificial intelligence, multimedia processing, or data analysis, may require updates to maintain their relevance and efficiency. When a system is highly maintainable, teams can adopt new frameworks, enhance security mechanisms, update third-party integrations, and introduce advanced features without large-scale refactoring. This flexibility ensures the platform remains modern, fully functional, and aligned with a future digital strategy that supports the livelihoods and development of local artisans.

2.2.2 Possible Architectural Patterns

Chapter 5

2.2.2.1 Architectural Patterns A

- *Chapter 2.2.2.1 – 2.2.2.2 shall discuss three (3) possible architectural patterns that can enable or help the developer to achieve the QOs/QAs identified in Chapter 2.2.1.*
- *For each architectural pattern:*
 - *Describe the architecture, its basic structure, characteristics and properties*
 - *Explain **the strategy used by this architecture to achieve the selected quality objectives** Explain the architecture's pros and cons*

2.2.2.2 Architectural Patterns B

XXXX

2.2.2.3 Architectural Patterns C

XXX

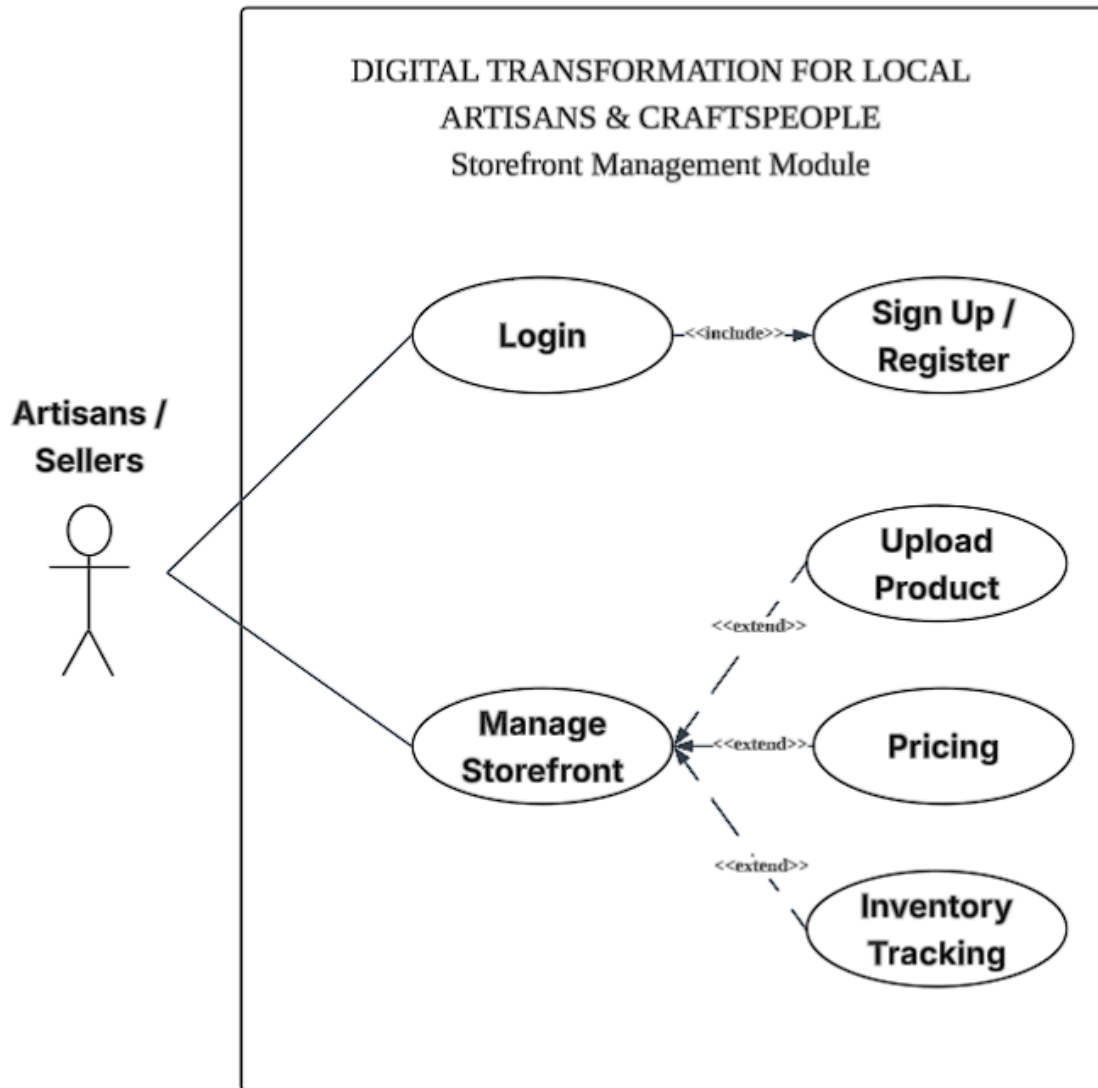
2.2.3 Selection Justification

- ***THIS IS A CRITICAL SECTION OF THIS CHAPTER.***
- *You are required to select your final architectural pattern from the architectural patterns discussed in Chapter 2.2.2.1 – 2.2.2.3*
- *The selection can either be one of the discussed patterns OR a combination of them*
- *Your discussion shall include:*
 - *Why did you choose the selected architectures over the others you explained?*
 - *If you combine multiple patterns into one, please justify the decision.*
 - *Your justifications must form the connection between the project and the architecture.*
 - *Don't explain general information the tutor/lecturer already knows – focus on your thought process and the features of your project.*

For Chapter 2 - describe your contents as precisely as possible, in point form or in table format. Do not write lengthy and unstructured paragraphs.

3. DETAILED DESIGN

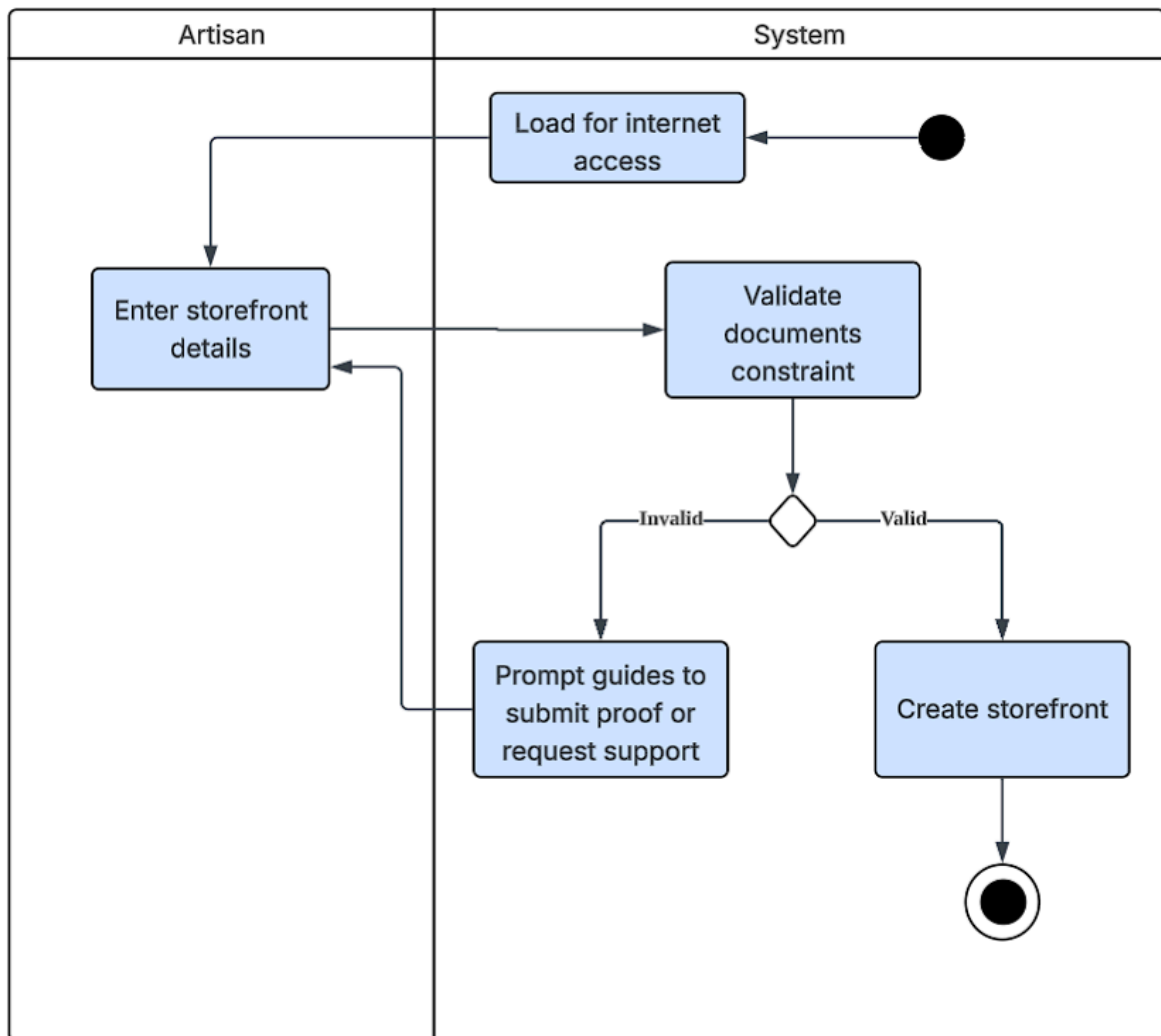
3.1 Module 1 : Storefront Management(Chang Kar Xi)



The diagram shown above is Digital Transformation for Local Artisans & Craftpeople's Storefront Management Module. Artisans are allowed to register a storefront for selling their products. After registration, artisans can log in to the storefront without filling up all the storefront details again. Storefront allows artisans to manage their products like upload product, pricing, and inventory tracking.

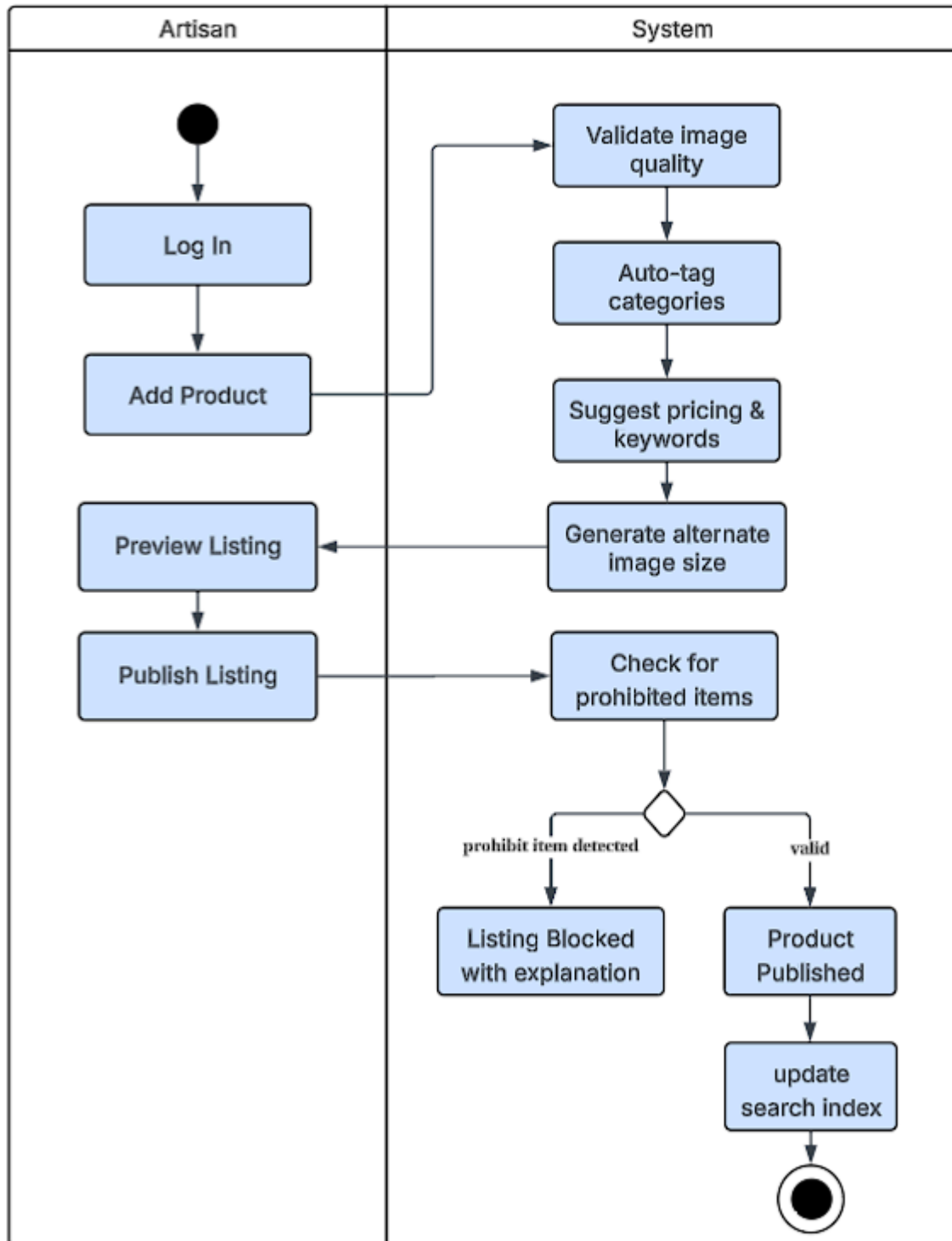
3.1.1 Activity Diagram

3.1.1.1 Activity Diagram for Use Case Artisan Onboarding & Storefront Setup



Paste your activity diagram here.

3.1.1.2 Activity Diagram for Use Case Upload Product Listing & Auto-Tag



Paste your activity diagram here.

**Repeat for other Use Case.*

3.1.2 Design Class Diagram

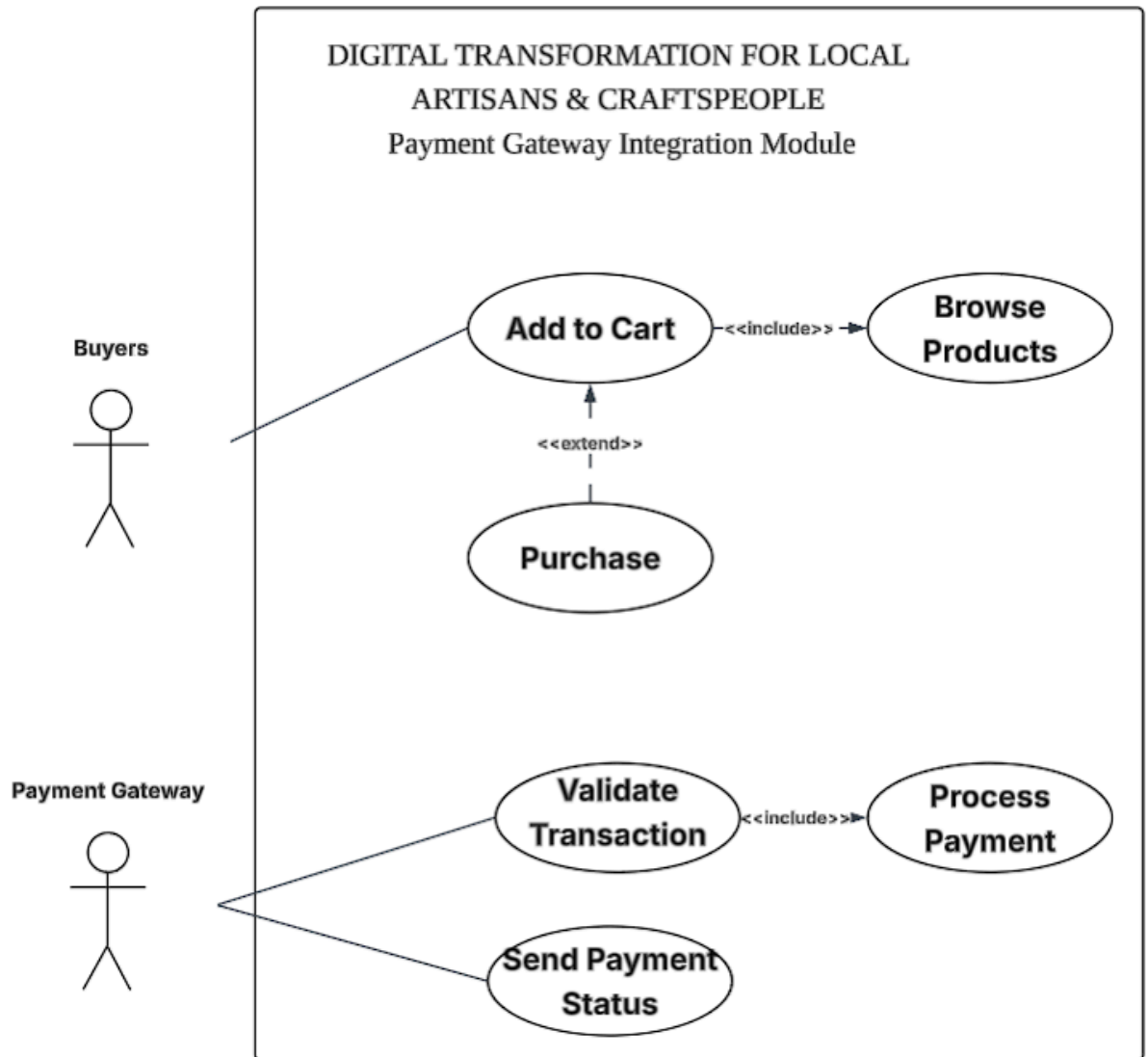
Note: draw a complete design class diagram which implements design pattern X in this section.

3.1.2.1 Rationale for Selected Design Pattern

Note:

- 1. Discuss the rationale for selecting the design pattern applied in 3.1.2, including critical issues and relevant trade/offs*
- 2. You may discuss other considered design patterns, provided you explain why you didn't choose them.*

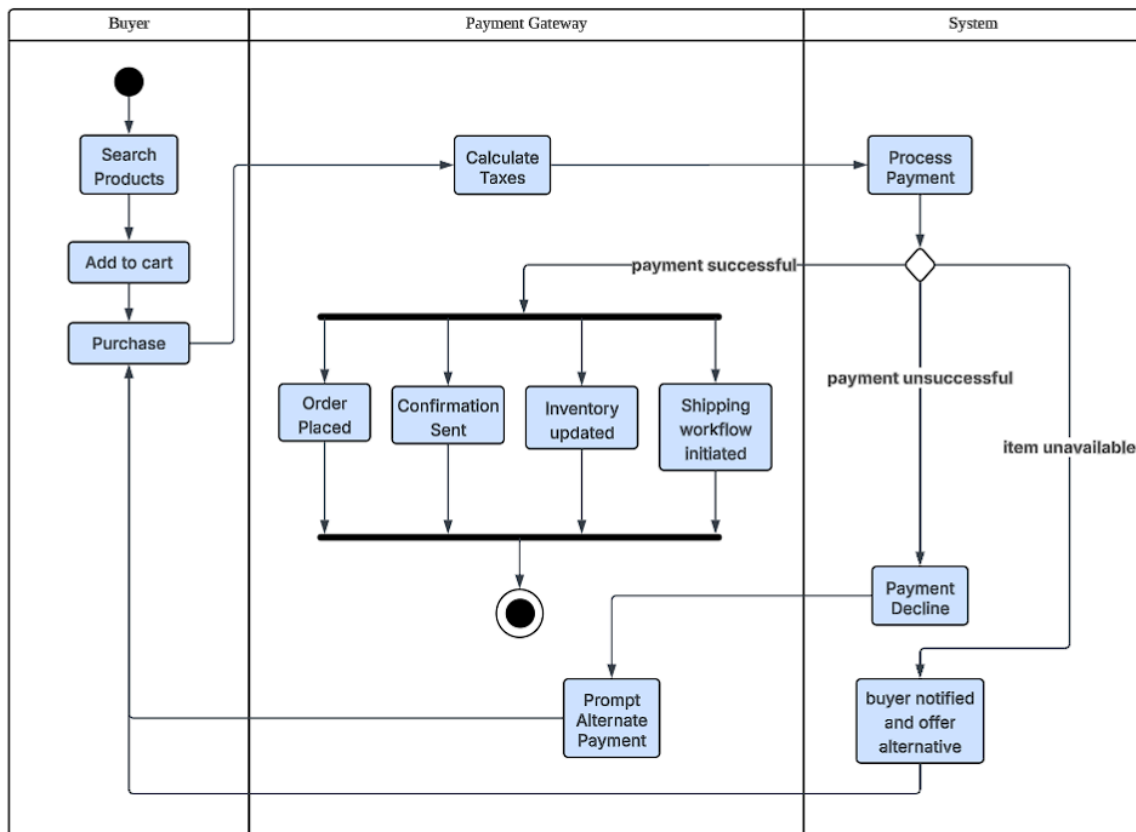
3.2 Module 2 : Payment Gateway Integration(Kennard Khoo Teng Phin)



The diagram above shows Digital Transformation for Local Artisans & Craftspeople 's Payment Gateway Integration Module. In this module, buyers are allowed to add products to the cart by browsing the products from artisan's storefront and purchase. Payment Gateway will process the payment while the buyer purchases the cart and validate the transaction. After successful validation, the payment gateway will send payment status back to the buyer for information.

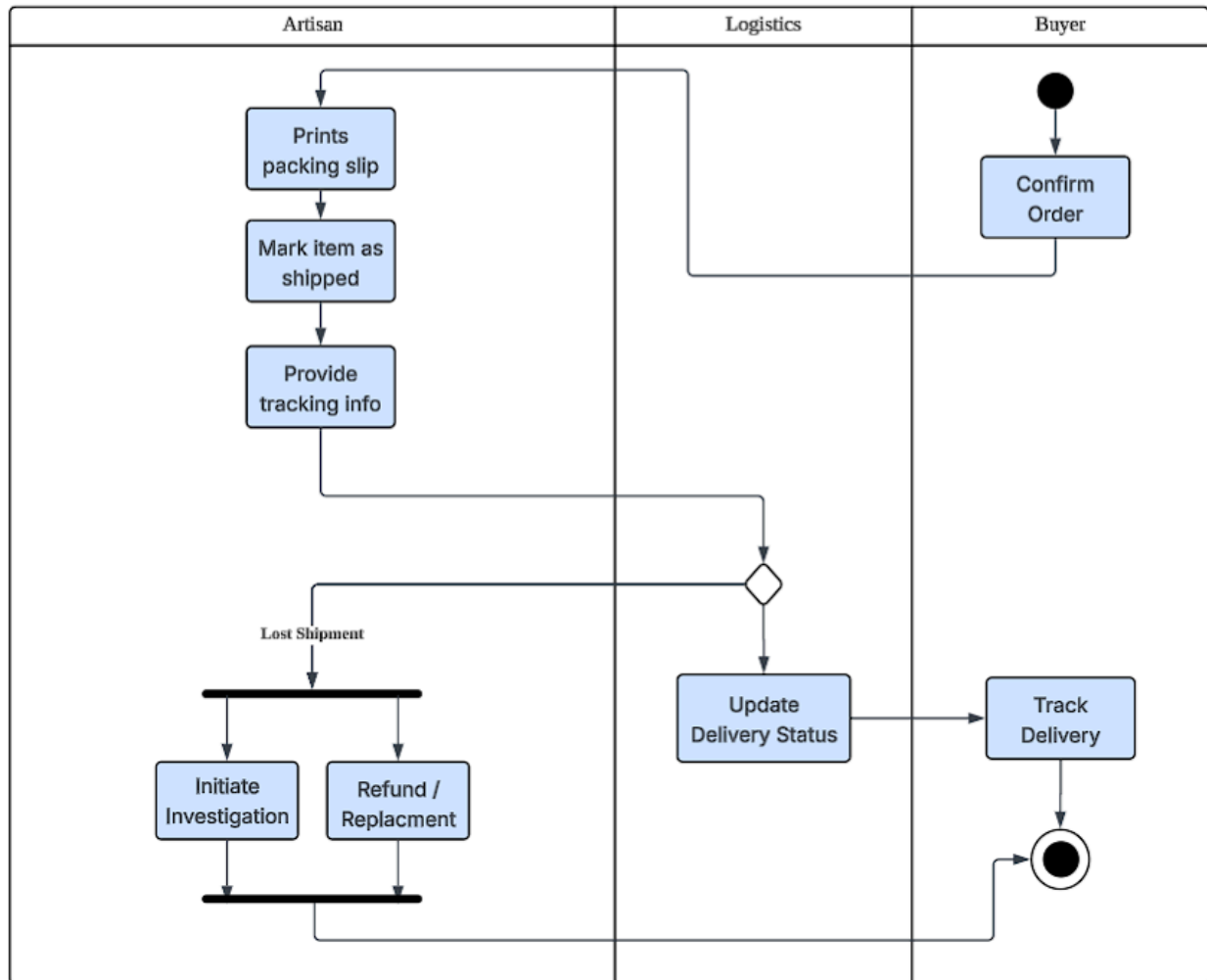
3.2.1 Activity Diagram

3.2.1.1 Activity Diagram for Use Case Buyer Search, Purchase & Payment



Paste your activity diagram here.

3.2.1.2 Activity Diagram for Use Case Order Fulfillment & Shipping



Paste your activity diagram here

**Repeat for other Use Case.*

3.2.2 Design Class Diagram

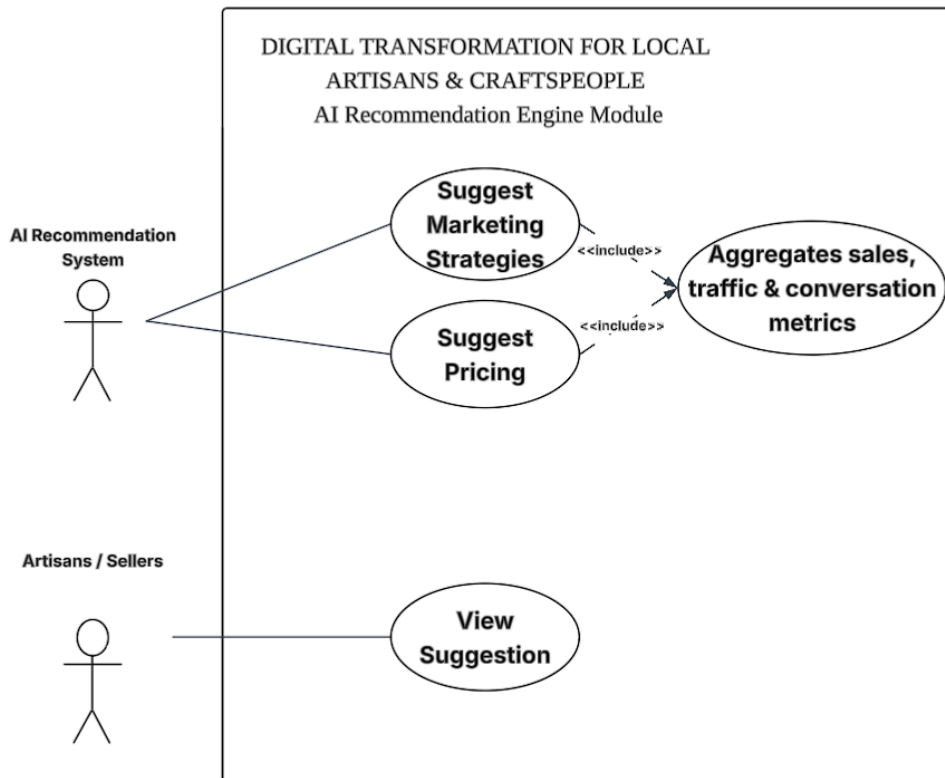
Note: draw a complete design class diagram which implements design pattern X in this section.

3.2.2.1 Rationale for Selected Design Pattern

Note:

- 1. Discuss the rationale for selecting the architecture described in 3.2.2, including critical issues and relevant trade-offs*
- 2. You may discuss other considered design patterns, provided you explain why you didn't choose them.*

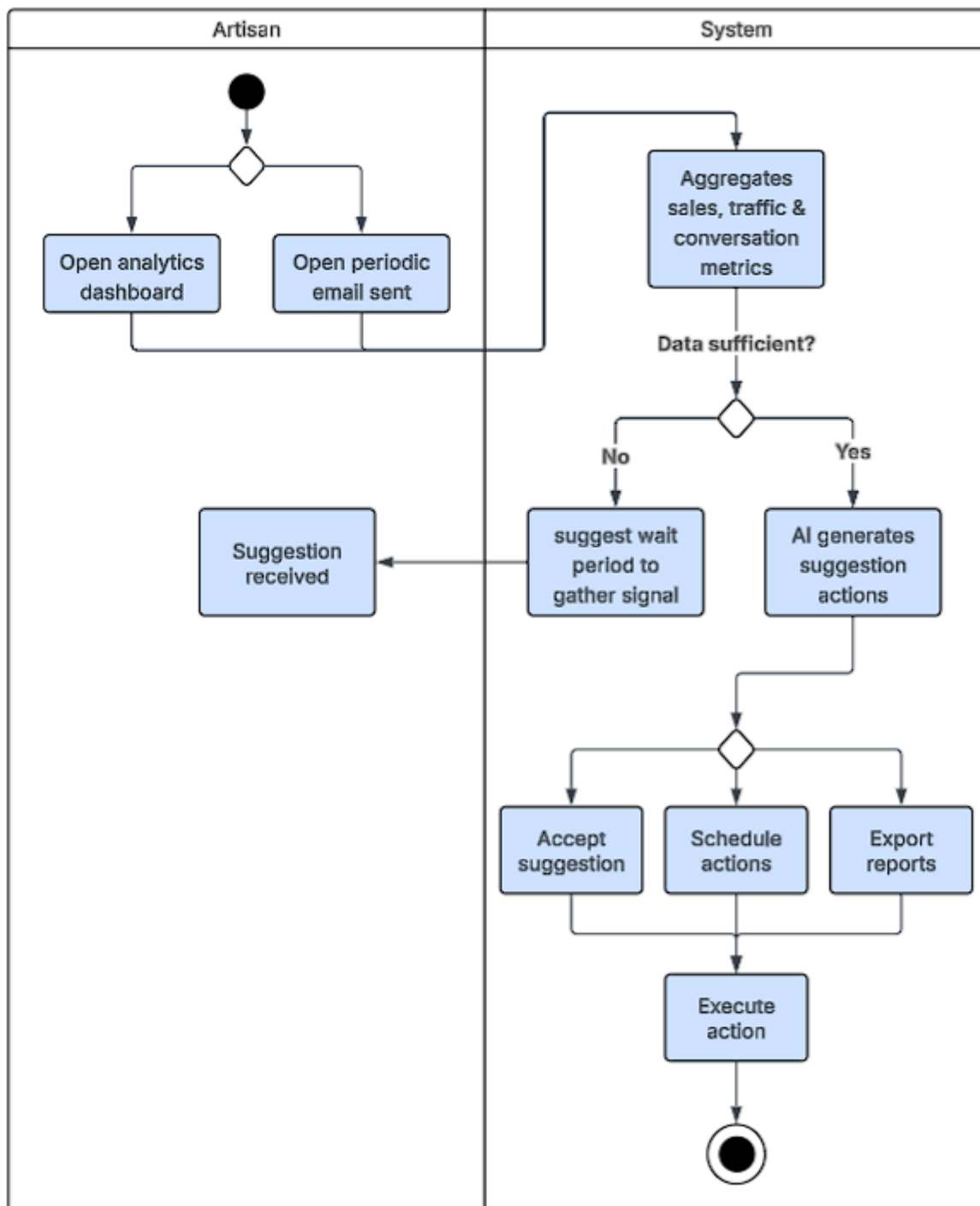
Note: Repeat 3.x for all team members.

3.3 Module 3 : AI Recommendation Engine(Tan Guan Hong)

The diagram above shows Digital Transformation for Local Artisans & Craftspeople 's AI Recommendation Engine Module. This module involves an AI Recommendation System to suggest marketing strategies and pricing to relevant artisans by aggregating sales, traffic and conversation metrics. Artisans are allowed to view the suggestion suggested by the AI Recommendation System to make further actions.

3.3.1 Activity Diagram

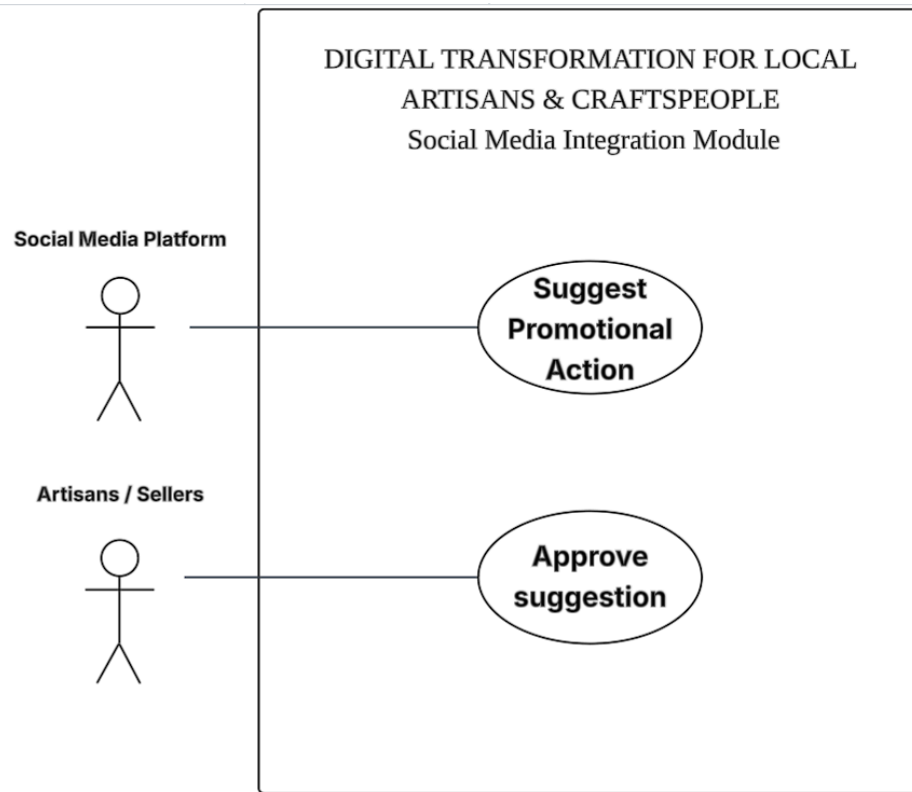
3.3.1.1 Activity Diagram for Use Case Sales Analytics & Recommendations



3.3.1.2 Activity Diagram for Use Case

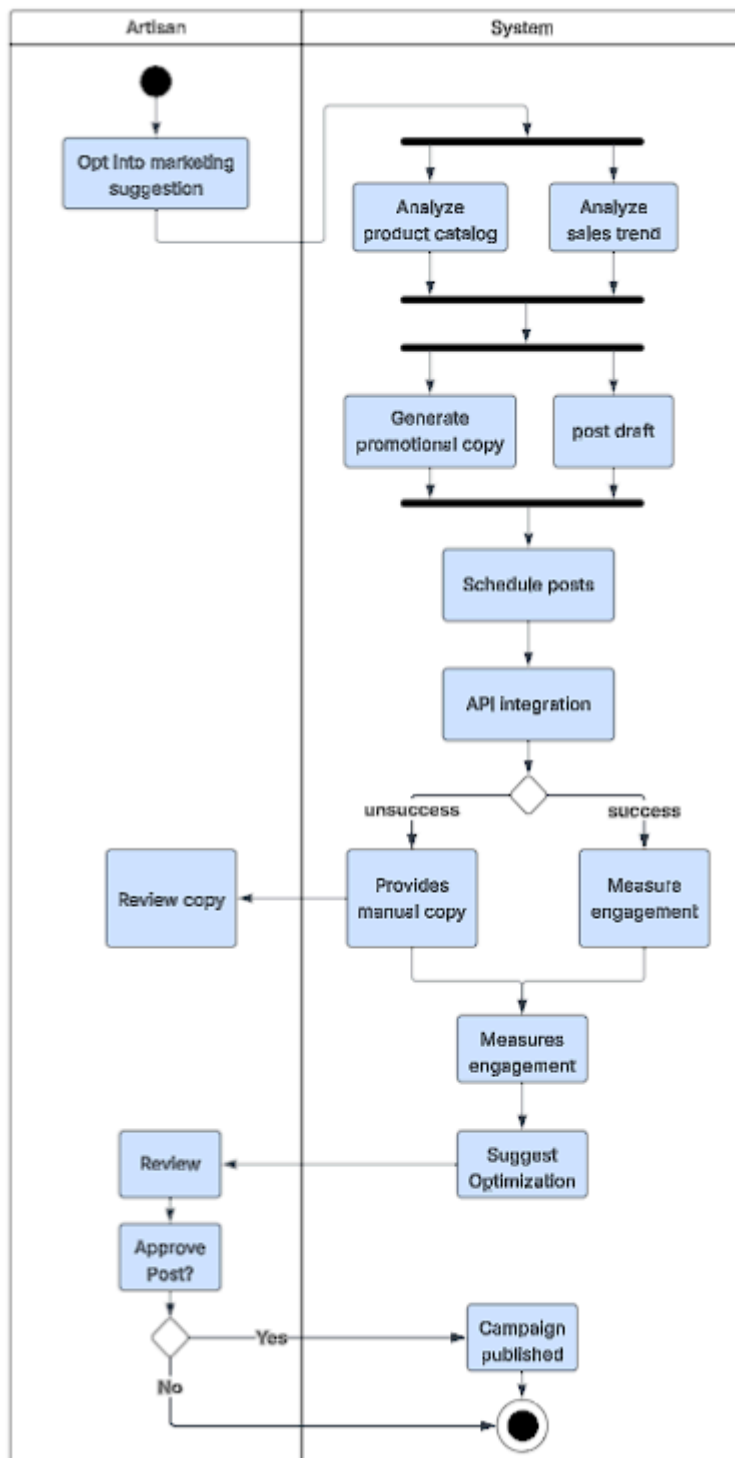
3.3.2 Design Class Diagram

3.3.2.1 Rationale for Selected Design Pattern

3.4 Module 4 : Social Media Integration(Ong Weng Kang)

3.4.1 Activity Diagram

3.4.1.1 Activity Diagram for Use Case AI Marketing & Social Integration



Paste your activity diagram here.

3.4.1.2 Activity Diagram for Use Case

Paste your activity diagram here.

**Repeat for other Use Case.*

3.4.2 Design Class Diagram

Note: draw a complete design class diagram which implements design pattern X in this section.

3.4.2.1 Rationale for Selected Design Pattern

Note:

3. *Discuss the rationale for selecting the design pattern applied in 3.1.2, including critical issues and relevant trade/offs*
4. *You may discuss other considered design patterns, provided you explain why you didn't choose them.*

4. USER INTERFACE / USER EXPERIENCE DESIGN

4.1 Module 1 : xxxxxx (1st student)

Note: place a minimum of 2 GUI / mock-up pages for each module.

4.2 Module 2 : xxxxxx (2nd student)

Note: place a minimum of 2 GUI / mock-up pages for each module.

NOTE :

- *Repeat 4.x for all team members.*
- *As one project, ALL the mock-up pages shall be consistent and cohesive*